NURSADUL MAMUN

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RESEARCH INTEREST

- Audio, Speech and Language Processing
- Speech Enhancement
- Auditory Nerve Modelling
- Cochlear Implants and Hearing aids
- Speech and Speaker Recognition
- Deep neural networks in speech recognition
- Speech Intelligibility and Quality measure
- Machine Learning and Artificial Intelligence

EDUCATIONAL QUALIFICATIONS:

PhD in Electrical and Electronic Engineering Erik Jonsson School of Engineering and Computer Science The University of Texas at Dallas, USA **Focus:** *Speech Enhancement for CI users.* **PhD Thesis Advisor**	May 2018 to Present (Pursuing) **Prof. Dr. John H. L. Hansen**
Master of Science in Biomedical Engineering University Malaya, Kuala Lumpur, Malaysia. **Focus:** *Speech Intelligibility prediction measure.* Master's Thesis **Master Thesis Advisor**	September 2013 to July 2015 'DISTINCTION' (Fully Research mode). Prediction Of Speech Intelligibility Using A Physiologically Based Model of The Auditory System. **Dr. Muhammad Shamsul Arefeen Zilany**
Bachelor of Science in Electrical and Electronic Engineering Chittagong University of Engineering and Technology, Bangladesh.	March 2008 to December 2012 (3.89 with 'HONORS' out of 4; 90.76%.)

Selected Courses:

- Speech and Speaker Recognition
- Probability, Random Processes and Statistics

ARTICLE IN ACADEMIC JOURNALS

2018

- **Nursadul Mamun**, S. A. Zilany. "A Spectrogram Orthogonal Polynomial Measure (SOPM) for the Prediction of Speech Intelligibility and Quality" Submitted to IEEE Transaction on Audio, Speech, and Language Processing.

2015

- [JP-3] **Nursadul Mamun**, W. A. Jassim, S. A. Zilany. "Prediction of Speech Intelligibility Using a Neurogram Orthogonal Polynomial Measure (NOPM)"). IEEE Transaction on Audio, Speech, and Language Processing, 2015 (ISI/SCOPUS *Cited Publication*), vol. 23(4), pp. 760-773.

2013

- [JP-2] **Nursadul Mamun**, Nahidul Haque Samrat, Choton Kanti Das, Sagor Ghosh."Design and Implementation of a Vehicle Monitoring System for Toll Collection". International Journal of Science & Engineering Research, 2013, V-4, Issue-4.
- [JP-1] Afrin Hossain, Tajrin Jahan Rumky, **Nursadul Mamun**."Implementation of smart energy meter with two way communication using GSM technology". International Journal of Science & Engineering Research, 2013, V-4, Issue-7.

CONFERENCE PRESENTATIONS

2018

- [CP-15] Hussnain Ali, **Nursadul Mamun**, Avamarie Bruggeman, Ram Charan Chandra Shekar, Juliana Saba, John H. L. Hansen "The CCi-MOBILE VOCODER" 176th Meeting Acoustical Society of America (ASA), 5-9 November 2018, Victoria, Canada**. (International)**
- [CP-14] Khadija Akter**, Nursadul Mamun** "Prediction of Speech Perception with Recovered Envelope Cues from TFS stimulus For Normal Hearing Listener" 4th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT 2018), 13-15 September 2018, MIST, Dhaka, Bangladesh**. (International)**
- [CP-13] **Nursadul Mamun**, Khadija Akter, Hussnain Ali, John H. L. Hansen "Measuring Speech Perception with Recovered Envelope Cues using the Peripheral Auditory Model" 176th Meeting Acoustical Society of America(ASA), 5-9 November 2018, Victoria, Canada**. (International)**
- [CP-12] Md. Saiful Islam**, Nursadul Mamun** and Muhammad S. Ullah "Speech Based Deception Detection Using Bispectral Analysis" 2nd International Conference on Graphics and Signal Processing (ICGSP), 6-8 October 2018, Sydney, Australia**. (International)**
- [CP-11] Antora Dev**, Nursadul Mamun** "Design of an EEG-based Brain Controlled Wheelchair for Quadriplegic Patients " IEEE 3rd International Conference for Convergence in Technology (I2CT), 6-8 April 2018, Pune, India**. (International)**

2017

- [CP-10] Eftekhar Hossain, **Nursadul Mamun** "Vehicle to Vehicle Communication Using RF and IR Technology" 2nd International Conference on Electrical & Electronic Engineering (ICEEE), 27-29 December 2017, RUET, Rajshahi, Bangladesh**. (International)**
- [CP-9] B. Patwary, Sabrina Abedin, Tasfia Tasbin and **Nursadul Mamun** "Conductivity Based Concentration Measurement Technique For Detecting Adulteration " Proceedings of the International Conference on Mechanical Engineering and Renewable Energy 2017 (ICMERE2017) 18 – 20 December, 2017, Chittagong, Bangladesh**. (International)**
- [CP-8] Taieba Taher, M. A. Haider, **Nursadul Mamun** "Smart Zone Sensing System with Accident Prevention" 2017 IEEE Region 10 Humanitarian Technology Conference (R10-HTC) 21 - 23 Dec 2017, Dhaka, Bangladesh. (International)
- [CP-7] Shoumya Chowdhury, **Nursadul Mamun**, A. A. Shahjamal Khan, Fahim Ahmed. "Text Dependent and Independent Speaker Recognition Using Neural Responses from the Model of the Auditory System" International Conference of Electrical, Electronic, and Communication Engineering, 2016 (EECE)**, 11-13** February 2017. (International)

2016

- [CP-6] Evelyn Davies-Venn**, Nursadul Mamun**, S. A. Zilany. "Computational auditory model of intensity effects on amplified speech perception" 39th Mid-Winter Meeting of the Association for Research in Otolaryngology (ARO), 20-24 Feb 2016, San Diego, California, USA (International).

2014

- [CP-5] S. A. Zilany**, Nursadul Mamun**, W. A. Jassim. "Prediction of Behavioral Speech Intelligibility

- and Quality using a Computational Model of the Auditory System" 37th Mid Winter Meeting of the Association for Research in Otolaryngology (ARO), 21-25 Feb 2014, Association for Research in Otolaryngology (ARO), (International).
- [CP-4] **Nursadul Mamun**, W. A. Jassim, S. A. Zilany. "Robust Gender Classification using Neural responses from the Model of the Auditory System" International Functional Electrical Simulation Society 2014 (IFESS)**,** 17 September 2014 to 19 September 2014, (International)
- [CP-3] **Nursadul Mamun**, W. A. Jassim, S. A. Zilany. "Speech-based Gender Classification Using Neurogram Orthogonal Polynomial measure**"** Net Regional Conference on Electrical Engineering (RCEE), 04 Mar 2014 to 05 Mar 2014, University Malaya and JICA Project for AUN/Speed-Net, (National)
- [CP-2] **Nursadul Mamun**, Nahidul Haque Samrat, Nur Muhammad Khan, Jahirul Islam."Multi-directional Solar Tracker Using Low Cost Photo Sensor Matrix". International Conference on Informatics, Electronics & Vision (ICIEV), IEEE, 2014.

  2013

- [CP-1] Nahidul Hoque Samrat, **Nursadul Mamun**, Shah Mukim Uddin, and Arif Islam. "Case study: Tidal current energy potential in Bangladesh." International Conference on Informatics, Electronics & Vision (ICIEV), pp. 1-5. IEEE, 2013.

HONOURS AND AWARDS:

Best Presentation Award: Best poster presenter in International Conference on Innovations in science, Innovation and Technology (ICISIT) held on 28-29 October 2016 in Bangladesh.

TEACHING EXPERIENCE:

Assistant Professor — 07 Feb. 2018- Present (Study Leave)
Lecturer — 24 April 2016 - 06 February 2018
Dept. of Electronics and Telecommunication Engineering, Chittagong University of Engineering and Technology (CUET), Bangladesh.

RESEARCH EXPERIENCE:

May 2018- Present
Research Assistant
Cochlear Implant Laboratory
Erik Jonsson School of Engineering and Computer Science
The University of Texas at Dallas

January 2013- February 2016.
Research Assistant
Auditory Neuroscience Laboratory
Department of Biomedical Engineering, University of Malaya, Malaysia.

PROFESSIONAL AFFILIATION/MEMBERSHIP

- Student member, IEEE.
- Student member, Acoustical Society of America (ASA).
- Professional member, Institute of Engineering of Bangladesh (IEB)

Technical Skill & Experience:

- Developed speech Intelligibility and quality prediction measure NOPM and SOPM.
- Experience with human auditory nerve model used for speaker/speech Identification and objective Intelligibility/quality measure.
- Experience with speech enhancement algorithms for cochlear implant users.
- Experience with vocoder system.
- Experience with CCi-Mobile research platform.
- Developed speech response-based gender classification system
- Experience with neural response-based speaker Identification system

Programming:

- **Programming Languages:** Python, C
- **Numeric Computation:** Matlab

References:

Dr. John H.L. Hansen, Professor
Electrical Engineering Department, The University of Texas at Dallas
Email: john.hansen@utdallas.edu